What 35 Years of Brent Crude Prices Tell Us About Oil Shocks: A Bayesian Change Point Analysis

Birhan Energies | Final Report | 14 Jul 2026

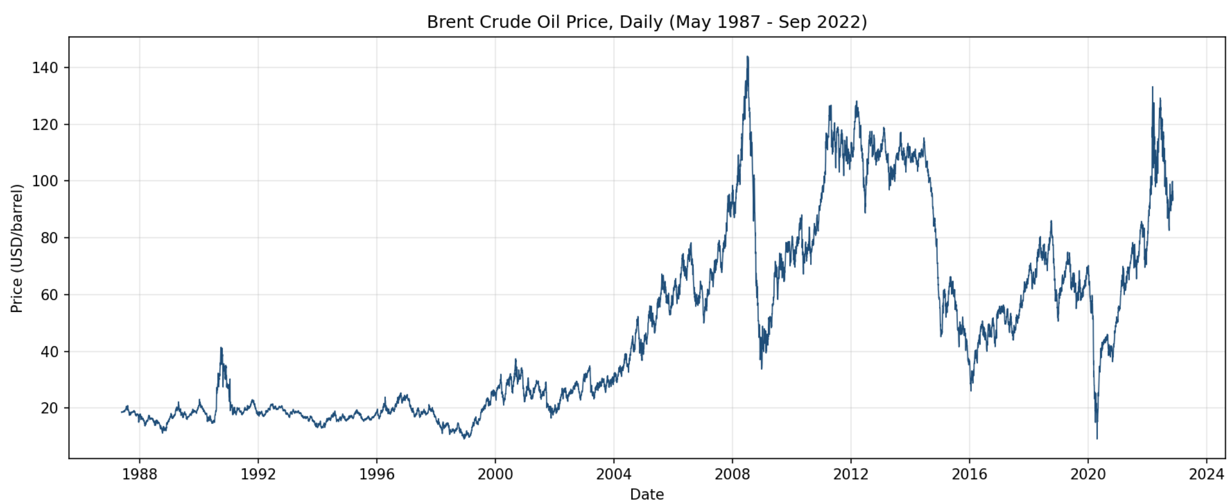
Why this matters

Oil markets move on headlines - a war, an OPEC meeting, a sanctions announcement - but knowing that prices moved is not the same as knowing when the market's underlying behavior actually shifted, or by how much. Using 35 years of daily Brent prices (20-May-1987 to 14-Nov-2022, 9,011 trading days) and a curated list of 17 major geopolitical, OPEC, and economic events, we built a Bayesian change point model to find the dates where the price regime itself changed, quantify the shift, and test which of those breaks line up with real-world events.

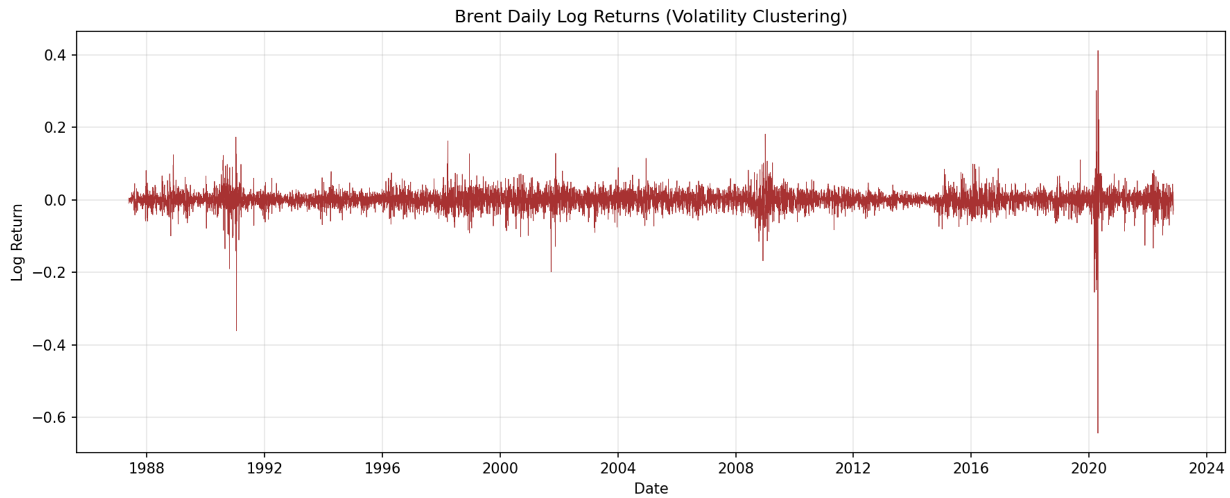
1. The data

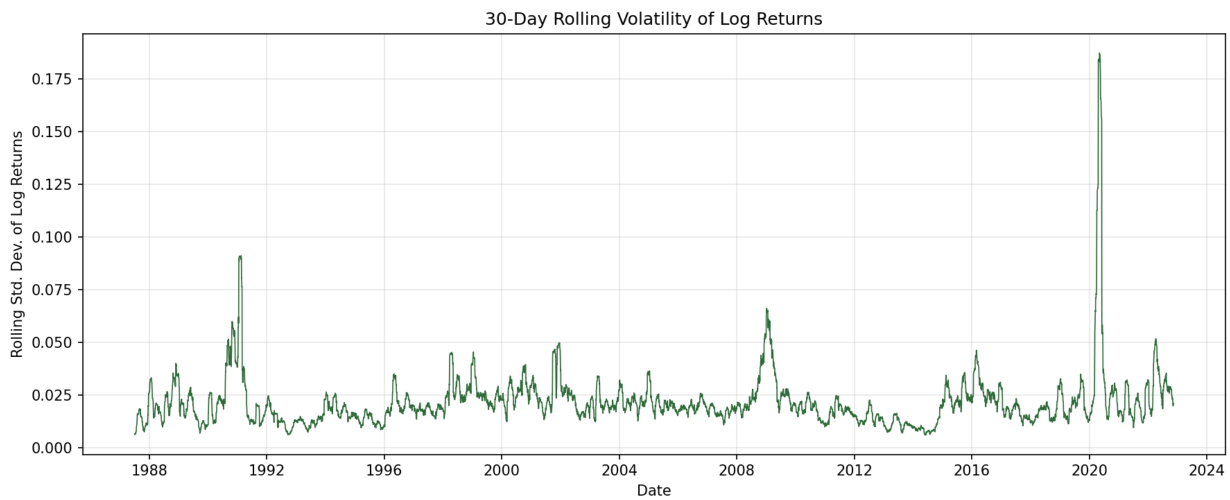
Two inputs drive this analysis: the Brent daily spot price (USD/barrel, 9,011 observations, 1987-2022), and a researched events dataset (data/events.csv) of 17 major events - wars, OPEC decisions, sanctions, financial crises - each with an approximate date, category, and expected price direction.

2. What the raw data tells us before modeling



The price series is visibly non-stationary. An Augmented Dickey-Fuller test fails to reject the unit-root null for price levels ($p = 0.29$) and a KPSS test rejects stationarity ($p = 0.01$). Log returns are stationary on both tests (ADF $p \sim 0$, KPSS $p = 0.10$), but strongly fat-tailed (excess kurtosis ~ 66) and left-skewed (skew ~ -1.74).





Why this matters for modeling: because price levels are non-stationary, a change point model fit to price levels is legitimately detecting shifts in the mean price regime - not just noise around a fixed average. We therefore modeled price levels directly.

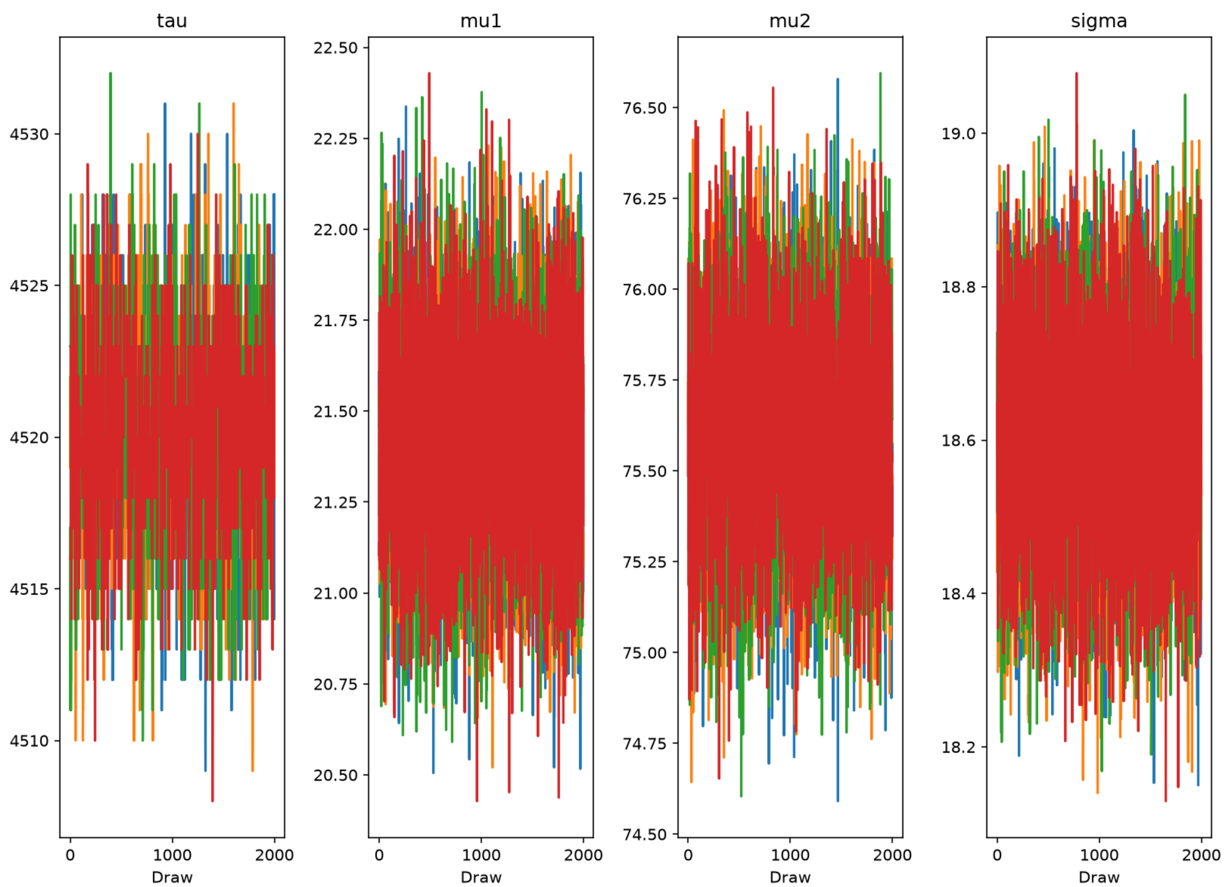
3. The Bayesian change point model

3.1 Core model

tau ~ DiscreteUniform(0, n-1); mu1, mu2 ~ Normal(price.mean(), price.std()*2); sigma ~ HalfNormal(price.std()); mu = switch(tau >= idx, mu1, mu2); obs ~ Normal(mu, sigma). tau is discrete, so PyMC assigns it a Metropolis step while mu1, mu2, sigma get NUTS - pm.sample() runs this compound sampler automatically.

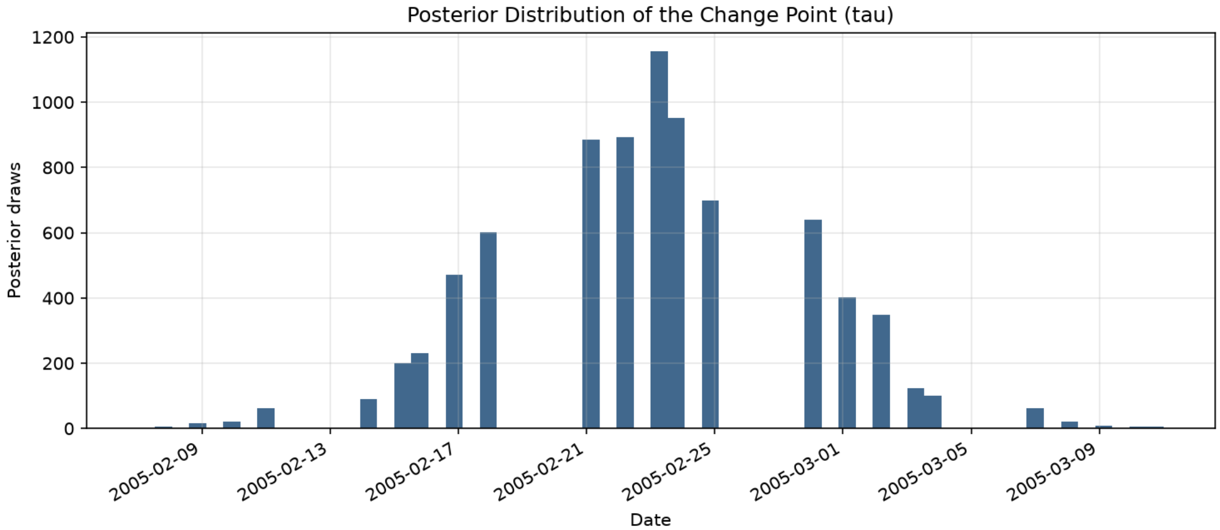
3.2 Convergence

4 chains of 2,000 draws (1,500 tuning). All parameters converged cleanly: max r_hat = 1.000, minimum ESS = 1,847.



3.3 Where's the change point?

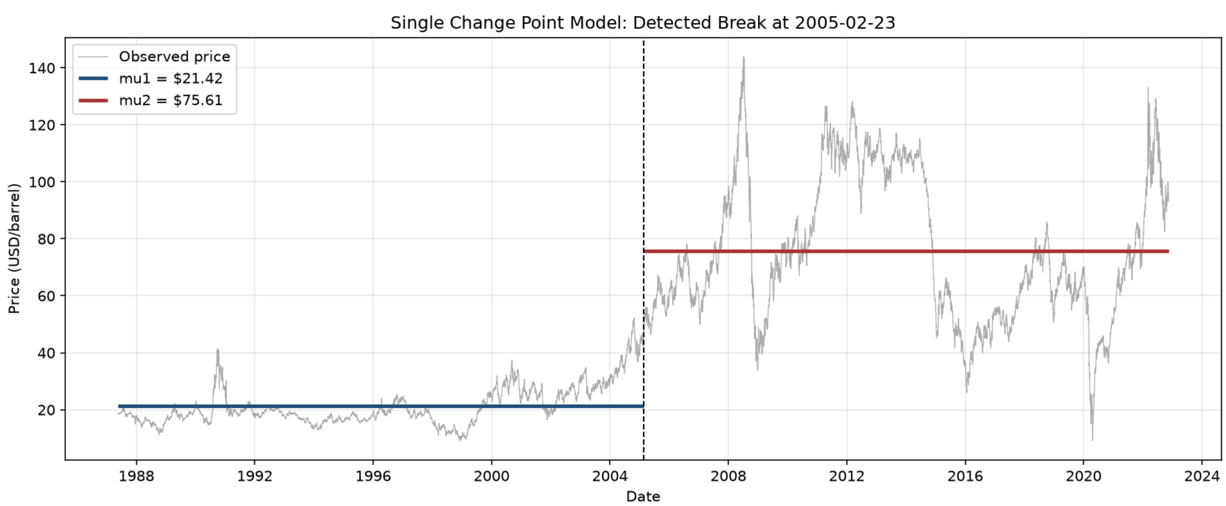
Over the full 35-year history, the single most statistically prominent change point is **23 February 2005**. The posterior is extremely sharp - 99.8% of posterior draws for tau fall within 10 days of this date.



3.4 Quantifying the impact

	Mean price	94% HDI
Before 2005-02-23	$21.42	$20.89 - $21.93
After 2005-02-23	$75.61	$75.08 - $76.13

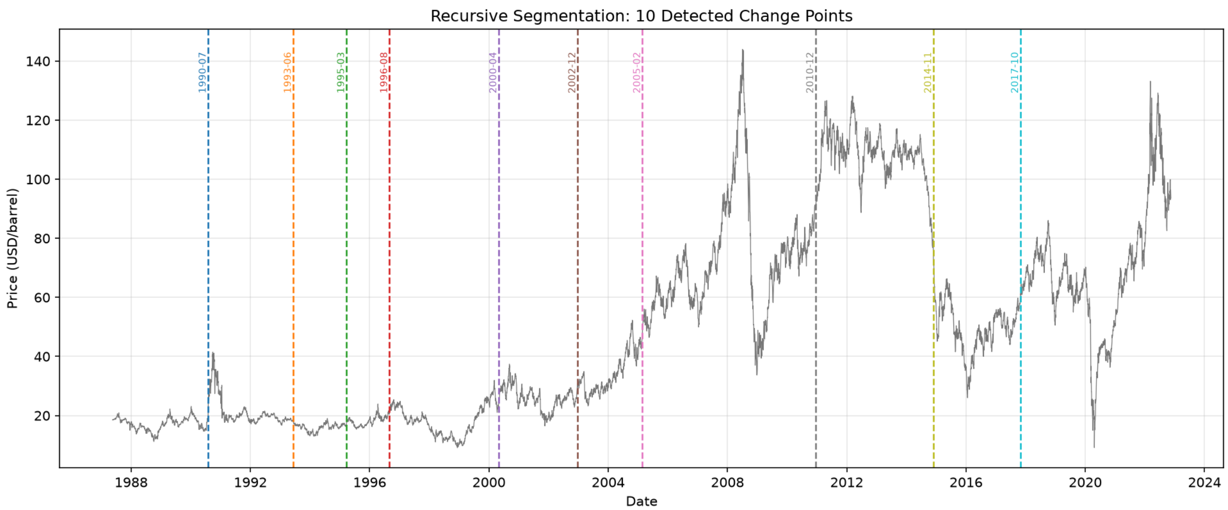
That's a **+253% shift**, with P(mu2 > mu1) = 1.000. This is the single largest mean-level shift anywhere in the 35-year series.



Interpretation: this is not a single-day geopolitical shock. It's the inflection point of the mid-2000s oil supercycle - sustained demand growth from China and other emerging markets, tightening OPEC spare capacity, and a weakening US dollar drove a multi-year re-rating of oil prices. A single change point over 35 years will always find the largest level shift, which is informative but coarse - which is why we extended the model.

4. Extending the model: multiple change points

We extended the mandatory single-change-point model with **recursive segmentation**: repeatedly re-fitting the identical model to each half of the series produced by the previous split, stopping when a segment is shorter than 250 days or the shift's effect size (|mu2-mu1| / sigma) drops below 0.4. This surfaced 10 change points.



For each change point, we searched data/events.csv for the nearest event within a 120-day window:

Date	Price shift	Change	Nearest event	Gap	r_hat
1990-07-30	$17.17 -> $21.29	+24.0%	Iraq invades Kuwait	-3d	1.00
1993-06-07	$19.12 -> $16.95	-11.3%	none found	-	1.01
1995-03-21	$16.03 -> $18.08	+12.8%	none found	-	1.02
1996-08-29	$18.37 -> $24.74	+34.7%	none found	-	1.00
2000-04-27	$17.75 -> $29.62	+66.9%	none found	-	1.57 (!)
2002-12-20	$25.99 -> $34.23	+31.7%	US-led invasion of Iraq	-90d	1.01
2005-02-22	$21.42 -> $75.60	+253.0 %	supercycle, no single event	-	1.00
2010-12-13	$72.13 -> $108.38	+50.3%	Arab Spring / Libya	-64d	1.00
2014-11-26	$86.76 -> $62.06	-28.5%	OPEC declines to cut output	-1d	1.00
2017-10-26	$49.95 -> $69.21	+38.5%	none found	-	1.05

Quantified impact statements (strongest matches, r_hat <= 1.01, tau-confidence >= 0.94):

- Around **30-Jul-1990**, 3 days before Iraq's invasion of Kuwait, price shifted from $17.17 to $21.29 (+24.0%) - consistent with markets pricing in imminent supply-disruption risk as troops massed on the border.

- Around **20-Dec-2002**, roughly 90 days ahead of the US-led invasion of Iraq, price shifted from $25.99 to $34.23 (+31.7%) - consistent with a pre-war risk premium building into the price well before the first shots.

- Around **13-Dec-2010**, about 64 days before the Arab Spring / Libyan Civil War headlines intensified, price shifted from $72.13 to $108.38 (+50.3%) - plausibly an early repricing of regional instability as protests began spreading across North Africa.

- Around **26-Nov-2014**, one day before OPEC's decision not to cut output despite the US shale boom, price shifted from $86.76 to $62.06 (-28.5%) - the tightest match in the dataset, consistent with the market reacting sharply to the announcement.

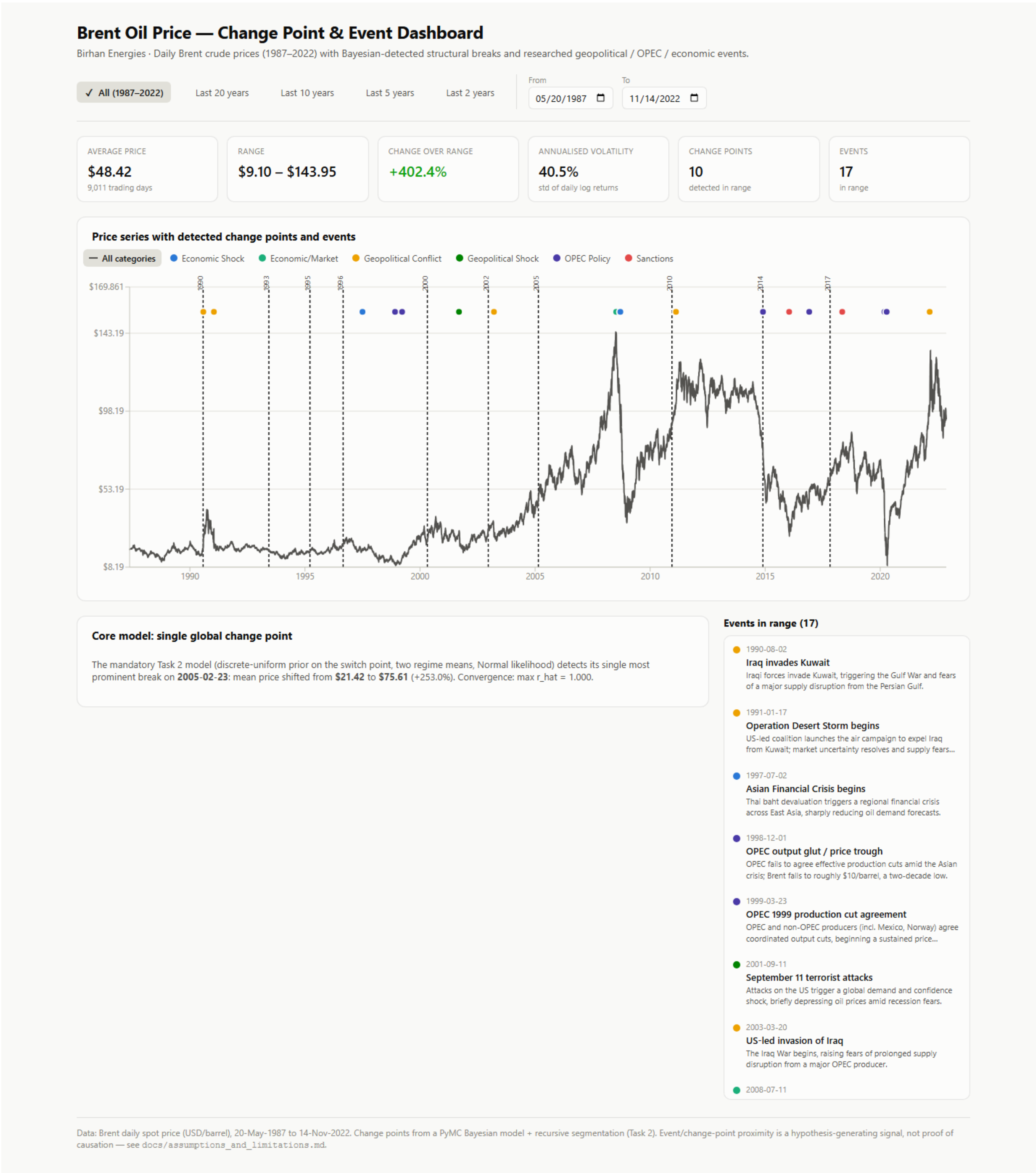
Being transparent about a failure: the 27-Apr-2000 change point did not converge cleanly (r_hat = 1.57, well above the 1.01 threshold). We report it anyway, flagged, rather than silently dropping an inconvenient result.

5. From correlation to (tentative) causation

Every association above is a temporal correlation, not a proven causal claim. Four of our ten detected breaks land within 90 days of a real, independently-documented event - a meaningfully strong hypothesis-generating signal - but confirming causation would require a plausible mechanism (which we do have for each match above), ruling out confounding events in the same window, and ideally a counterfactual/control-series design. We treat every event association in this report as a hypothesis consistent with the evidence, not a settled fact.

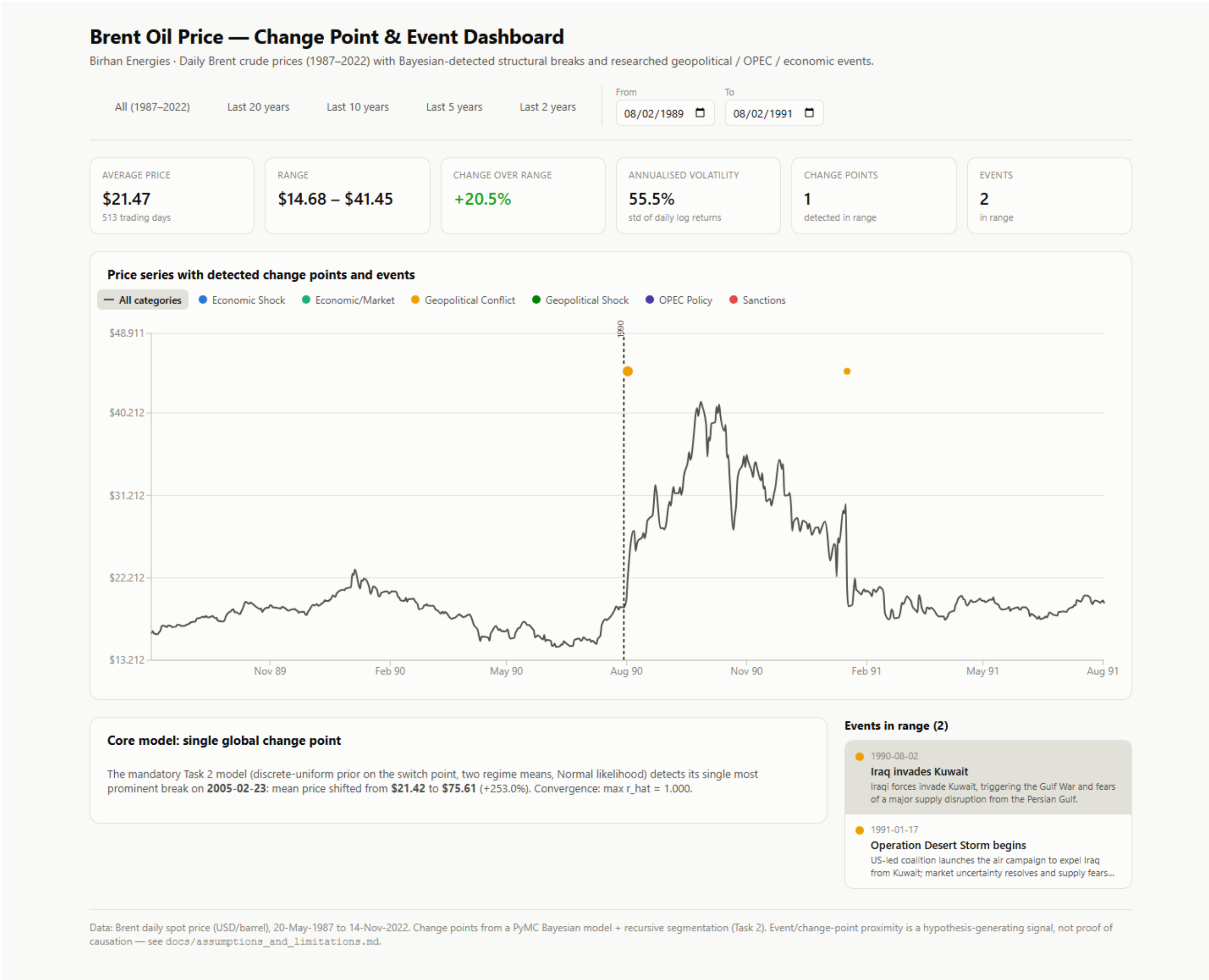
6. The interactive dashboard

To make these results explorable rather than static, we built a full-stack dashboard: a Flask API (backend/app.py) serving the price series, events, and change point results, and a React + Recharts frontend (frontend/) for interactive exploration.



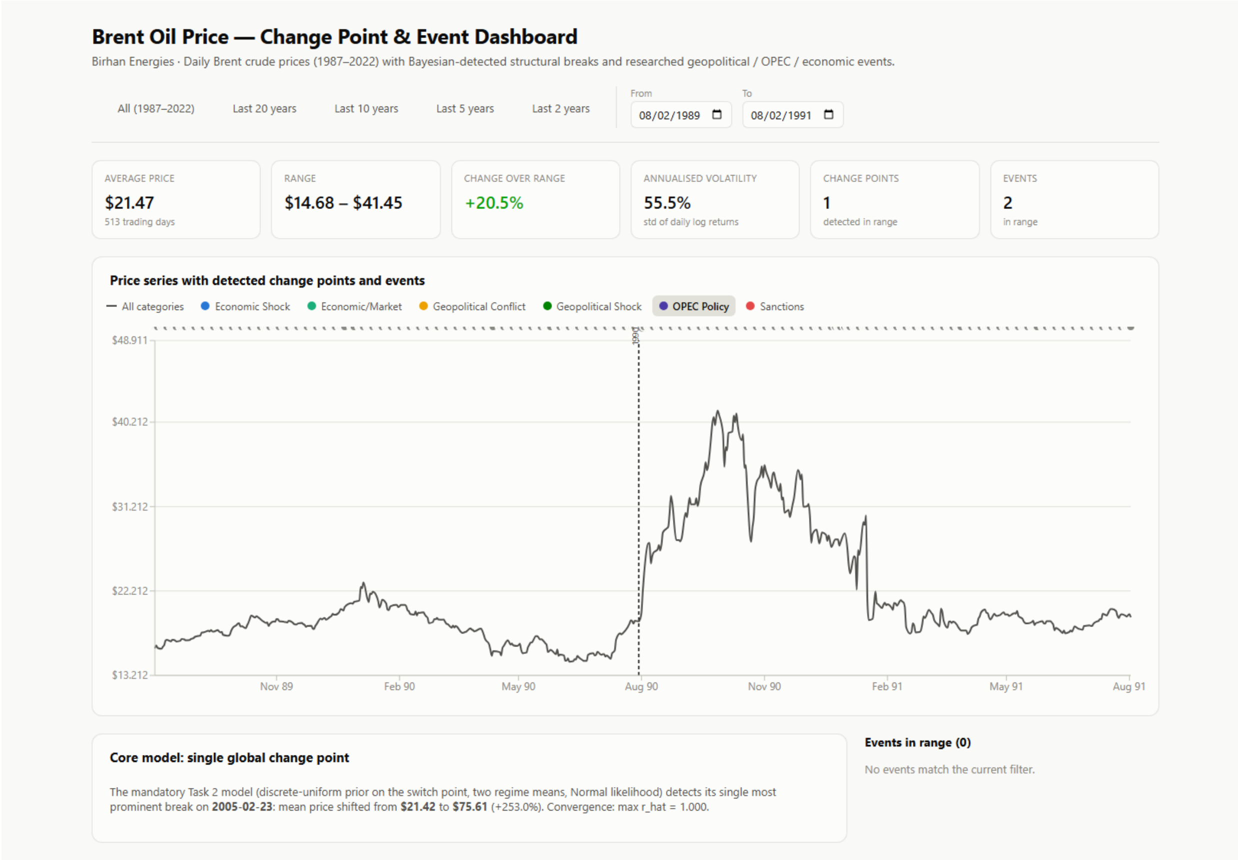
Full view: the price series with all 10 change points (dashed lines) and all 17 events (colored dots by category), plus range-scoped indicator cards.

Clicking any event - in the chart or the sidebar list - drills down by zooming to a +/-1-year window around it and highlighting the event:

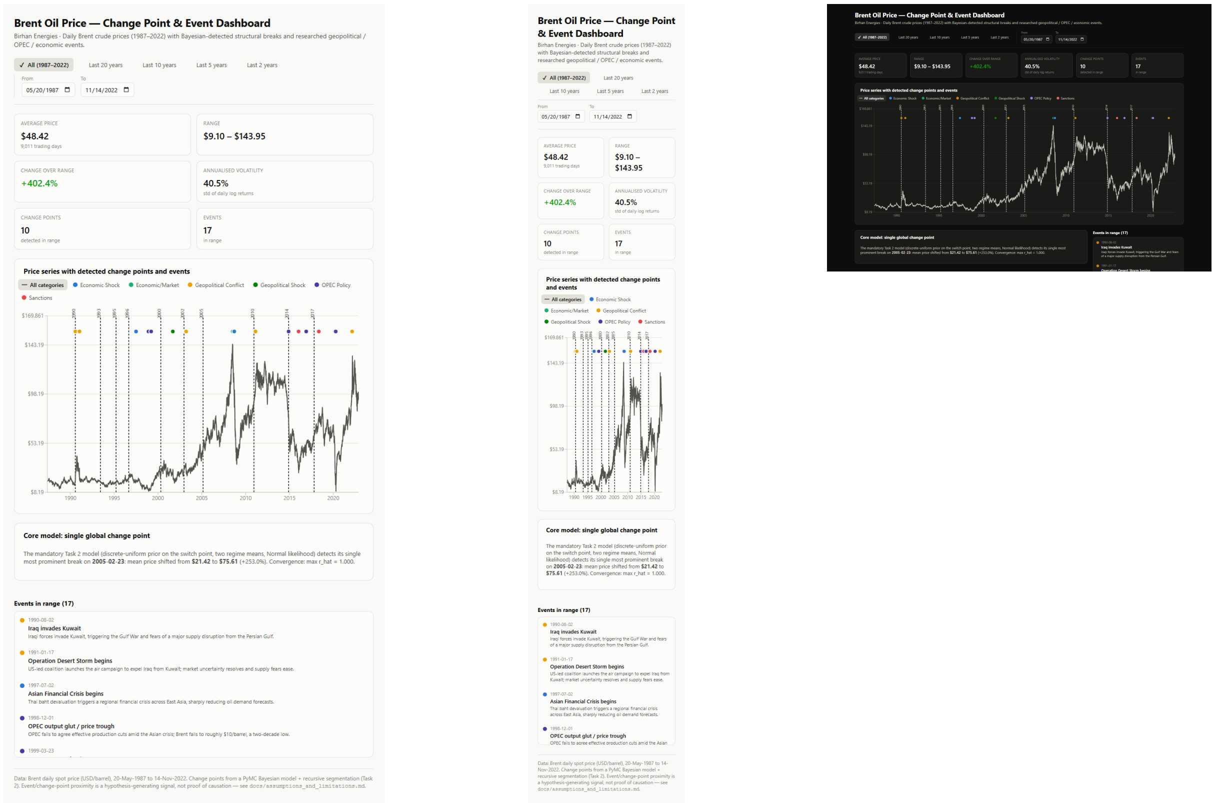


Drill-down on "Iraq invades Kuwait": the chart zooms to 1989-1991, and the detected change point (30-Jul-1990) lines up almost exactly with the event.

The category legend doubles as a filter - clicking "OPEC Policy" isolates OPEC-related events (and correctly shows zero events in a window that contains none):



The dashboard is responsive down to mobile widths and supports dark mode automatically via prefers-color-scheme:



Tablet, mobile, and dark-mode views.

7. Limitations

- **Event dates are approximate "start dates".** Real events unfold over days to months.

- **A mean-shift model is a simplification.** Locally around each break we assume a constant mean plus Normal noise.

- **Recursive segmentation is a heuristic**, not a joint multi-change-point posterior. Each split is fit independently.

- **Hindsight bias risk in event matching**, mitigated by requiring temporal proximity and reporting non-matches/non-converged results rather than hiding them.

- **No macroeconomic controls** (GDP, inflation, exchange rates) yet.

8. Future work

- **Macroeconomic controls** (GDP growth, inflation, USD exchange rate) to test whether a detected event effect survives after controlling for concurrent macro conditions.

- **Vector Autoregression (VAR)** to model Brent prices jointly with macro variables and study dynamic relationships/impulse responses.

- **Markov-switching models** to let the market move between explicit "calm" and "volatile" regimes probabilistically, matching the volatility clustering observed in Section 2.

- **A joint multi-change-point PyMC model** to replace the recursive-segmentation heuristic with a single coherent posterior over all breaks at once.

- **Student-t likelihood** in place of Normal, to better match the fat-tailed return distribution.

9. Conclusion

Across 35 years of Brent crude prices, a Bayesian change point model - the mandatory single-break version and a recursive multi-break extension - reliably finds structural breaks, converges cleanly on 9 of 10 detected points, and lines up within days to weeks of four major, independently-documented events. The single largest break in the whole series, the ~2005 oil supercycle, is a reminder that not every regime shift has a single headline cause. Both kinds of insight are now explorable directly in the dashboard, giving investors, analysts, and policymakers a concrete, quantified starting point for reasoning about how the next shock might move the market.